

ALFAAL13356

## N-tert-Butyl-2-benzothiazolesulfenamide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明:  
Product Description: N-叔丁基-2-苯并噻唑次磺酰胺  
N-tert-Butyl-2-benzothiazolesulfenamide

Cat No. : L13356  
CAS No 95-31-8  
Molecular Formula C11 H14 N2 S2

Supplier Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
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Emergency Telephone Number For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
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**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

Physical State  
Solid

Appearance  
White - Cream

Odor  
No information available

#### Emergency Overview

May cause an allergic skin reaction. Very toxic to aquatic life with long lasting effects.

#### Classification of the substance or mixture

|                          |             |
|--------------------------|-------------|
| Skin Sensitization       | Category 1B |
| Acute aquatic toxicity   | Category 1  |
| Chronic aquatic toxicity | Category 1  |

#### Label Elements



Signal Word

Warning

Hazard Statements

# SAFETY DATA SHEET

## N-tert-Butyl-2-benzothiazolesulfenamide

H317 - May cause an allergic skin reaction  
H410 - Very toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves

#### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 - Store in a well-ventilated place

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Physical and Chemical Hazards

None identified.

### Health Hazards

May cause an allergic skin reaction.

### Environmental hazards

Very toxic to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil. The product is insoluble and sinks in water.

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                               | CAS No  | Weight % |
|-----------------------------------------|---------|----------|
| N-tert-Butyl-2-benzothiazolesulfenamide | 95-31-8 | <=100    |

## SECTION 4. FIRST AID MEASURES

### General Advice

If symptoms persist, call a physician.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

### Most important symptoms and effects

None reasonably foreseeable. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### Self-Protection of the First Aider

No special precautions required.

### Notes to Physician

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

**Storage**

Keep container tightly closed in a dry and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

None under normal use conditions. .

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## N-tert-Butyl-2-benzothiazolesulfenamide

### Personal protective equipment

|                        |                                                                                 |
|------------------------|---------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Wear safety glasses with side shields (or goggles) (European standard - EN 166) |
| <b>Hand Protection</b> | Protective gloves                                                               |

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                   |                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>   | Long sleeved clothing                                                                                                                                                                                  |
| <b>Respiratory Protection</b>     | No protective equipment is needed under normal use conditions.                                                                                                                                         |
| <b>Large scale/emergency use</b>  | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particle filter |
| <b>Small scale/Laboratory use</b> | Maintain adequate ventilation                                                                                                                                                                          |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                             |                                          |
|------------------------------------------------|-----------------------------|------------------------------------------|
| <b>Appearance</b>                              | White - Cream               |                                          |
| <b>Physical State</b>                          | Solid                       |                                          |
| <b>Odor</b>                                    | No information available    |                                          |
| <b>Odor Threshold</b>                          | No data available           |                                          |
| <b>pH</b>                                      | Not applicable              |                                          |
| <b>Melting Point/Range</b>                     | 105 - 110 °C / 221 - 230 °F |                                          |
| <b>Softening Point</b>                         | No data available           |                                          |
| <b>Boiling Point/Range</b>                     | No information available    |                                          |
| <b>Flash Point</b>                             | 165 °C / 329 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable              | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available    |                                          |
| <b>Explosion Limits</b>                        | No data available           |                                          |
| <b>Vapor Pressure</b>                          | No data available           |                                          |
| <b>Vapor Density</b>                           | Not applicable              | Solid                                    |
| <b>Specific Gravity / Density</b>              | 1.29 g/cm3                  | @ 20 °C                                  |
| <b>Bulk Density</b>                            | No data available           |                                          |
| <b>Water Solubility</b>                        | Insoluble in water          |                                          |
| <b>Solubility in other solvents</b>            | No information available    |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                             |                                          |
| <b>Component</b>                               | <b>log Pow</b>              |                                          |
| N-tert-Butyl-2-benzothiazolesulfenamid 3.36    |                             |                                          |

# SAFETY DATA SHEET

## N-tert-Butyl-2-benzothiazolesulfenamide

|                           |                          |       |
|---------------------------|--------------------------|-------|
| e                         |                          |       |
| Autoignition Temperature  | No data available        |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |
| Molecular Formula         | C11 H14 N2 S2            |       |
| Molecular Weight          | 238.38                   |       |

### SECTION 10. STABILITY AND REACTIVITY

|                          |                                 |
|--------------------------|---------------------------------|
| Stability                | Stable under normal conditions. |
| Hazardous Reactions      | None under normal processing.   |
| Hazardous Polymerization | No information available.       |
| Conditions to Avoid      | None known.                     |
| Materials to avoid       | Oxidizing agent.                |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides.

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

##### (a) acute toxicity;

| Component                               | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation |
|-----------------------------------------|---------------------------|------------------------------|-----------------|
| N-tert-Butyl-2-benzothiazolesulfenamide | LD50 = 6850 mg/kg ( Rat ) | LD50 > 7940 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

|             |                                         |
|-------------|-----------------------------------------|
| Respiratory | No data available                       |
| Skin        | Sub-category 1B                         |
|             | May cause sensitization by skin contact |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

# SAFETY DATA SHEET

## N-tert-Butyl-2-benzothiazolesulfenamide

**Target Organs** No information available.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                               | Freshwater Fish                                                                                                                                                | Water Flea                            | Freshwater Algae                                        | Microtox |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------|----------|
| N-tert-Butyl-2-benzothiazolesulfenamide | LC50: > 0.3 mg/L, 96h static (Pimephales promelas)<br>LC50: > 0.3 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: > 0.3 mg/L, 96h static (Lepomis macrochirus) | EC50: > 0.3 mg/L, 48h (Daphnia magna) | EC50: > 0.3 mg/L, 96h (Pseudokirchneriella subcapitata) |          |

#### Persistence and Degradability

##### Persistence

May persist.

##### Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

#### Bioaccumulative Potential

Product has a high potential to bioconcentrate

| Component                               | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------------|---------|-------------------------------|
| N-tert-Butyl-2-benzothiazolesulfenamide | 3.36    | No data available             |

#### Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and sinks in water Is not likely mobile in the environment due its low water solubility Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

#### Persistent Organic Pollutant

This product does not contain any known or suspected substance

#### Ozone Depletion Potential

This product does not contain any known or suspected substance

### SECTION 13. DISPOSAL CONSIDERATIONS

#### Waste from Residues/Unused Products

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

### SECTION 14. TRANSPORT INFORMATION

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## N-tert-Butyl-2-benzothiazolesulfenamide

### Road and Rail Transport

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (N-tert-Butyl-2-benzothiazolesulfenamide)  
Hazard Class 9  
Packing Group III

### IMDG/IMO

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (N-tert-Butyl-2-benzothiazolesulfenamide)  
Hazard Class 9  
Packing Group III

### IATA

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (N-tert-Butyl-2-benzothiazolesulfenamide)  
Hazard Class 9  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS).

| Component                               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| N-tert-Butyl-2-benzothiazolesulfenamide | -                                                   | -                                       | X    | X     | 202-409-1 | X    | X   | X     | X    | X    | X    | KE-11372 |

### National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Revision Date 02-May-2024  
Revision Summary New emergency telephone response service provider.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Chemical incident response training.

# SAFETY DATA SHEET

## N-tert-Butyl-2-benzothiazolesulfenamide

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### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**